

Transfer learning

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Similarity and normalisation

Without opening the old lesson, explain one key idea from Similarity and normalisation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Sequence representations

Without opening the old lesson, explain one key idea from Sequence representations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Pooling and spatial information

Without opening the old lesson, explain one key idea from Pooling and spatial information. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Fit and compare for Transfer learning. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Transfer learning. Proposed explanation: “Frozen means the whole model cannot learn”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Fit a new head independently

A separate frozen extractor is $h = 2 \cdot x_1 + x_2$. Training inputs $[1, 0]$, $[1, 2]$, $[2, 1]$, $[3, 1]$ have labels 0, 0, 1, 1. Compare thresholds 2, 4 and 6 using prediction 1 when $h \geq \text{threshold}$; choose the fewest mistakes, breaking ties toward the smaller threshold. Score it on validation $[2, 0]$ with label 1 and $[0, 2]$ with label 0. Compare with the training-majority baseline, whose tie rule chooses 0.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Trace the reused pipeline

For input $[4, 2]$, compute h using the original extractor and the class at threshold 5. Identify the frozen weights and bias, the fitted quantity, and why h is not a probability.

My reasoning and evidence

6. W2 Fit and compare

For the four target training records, give the number of mistakes at thresholds 3, 5 and 7 and choose the fitted threshold. Then score that head and the declared constant baseline on V1 and V2. Explain which

data fitted the head.

My reasoning and evidence

7. W3 Diagnose a lost distinction

For the first-count-greater target, explain why the original extractor cannot distinguish U [1, 3] from V [3, 1]. Trace the hand-chosen [1, -1] extractor and threshold 1. Does this show that fine tuning has been trained and independently validated?

My reasoning and evidence

8. W4 Fit a new head independently

A separate frozen extractor is $h = 2 \cdot x_1 + x_2$. Training inputs [1, 0], [1, 2], [2, 1], [3, 1] have labels 0, 0, 1, 1. Compare thresholds 2, 4 and 6 using prediction 1 when $h \geq \text{threshold}$; choose the fewest mistakes, breaking ties toward the smaller threshold. Score it on validation [2, 0] with label 1 and [0, 2] with label 0. Compare with the training-majority baseline, whose tie rule chooses 0.

My reasoning and evidence

9. W5 Plan credible evidence

A report omits the source checkpoint and preprocessing, claims faster learning without timing, and repeatedly chooses settings using the final evaluation set. Name the missing records and propose a fair comparison with a model trained from scratch.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled transfer learning evidence sheet to Image-pattern Network Investigation. Show how you can trace a reused feature extractor and a newly fitted output head. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Transfer learning

Transfer a fixed feature extractor while identifying what is actually refitted.

Frozen extractor $h = \text{left} + 2 \times \text{right}$. New-task training inputs $[1, 0] \text{label} 0, [2, 1] \text{label} 0, [1, 3] \text{label} 1$. Class 1 when $h \geq \text{threshold}$; candidates 3 or 5.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Transfer learning

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Extract features

Training values are 1, 4, 7.



2. Fit new head

Threshold 3 makes 1 mistake; threshold 5 makes 0, so choose 5.



3. Check held-out input

Query [2, 2] has $h = 6$, so the head predicts class 1; its reference is not supplied.

CHANGE ONE THING

Query changes to [2, 1].

Explain the changed result using the matching lesson.

Which part was trained for the new task? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Transfer a fixed feature extractor while identifying what is actually refitted.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Query [2, 2] has $h = 6$, so the head predicts class 1; its reference is not supplied.

1. Check held-out input
2. Extract features
3. Fit new head

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Check held-out input Query [2, 2] has $h = 6$, so the head predicts class 1; its reference is not supplied. Transfer learning can keep a representation fixed while fitting a smaller output head for a new task. Here the extractor always applies the same weights, while new labels choose the head's threshold. Applying the chosen head to a query is inference. State which part was frozen and which was fitted; this arithmetic does not establish the extractor is useful for every task.

Practice 2: explain the deciding evidence

Changed-case practice: Query changes to [2, 1]. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. Its h becomes 4 and the fitted head predicts class 0.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Its h becomes 4 and the fitted head predicts class 0. Extractor stays fixed; the changed input crosses the chosen head's boundary.

Repair a missing building block

Review Similarity and normalisation (ai-5/27). Goal: Calculate vector length and normalise a nonzero vector. Short practice: Why do the original two metrics select different winners? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Sequence representations (ai-5/25). Goal: Encode an ordered sequence with an explicit vocabulary. Short practice: Which record lets the program distinguish the first and last zeros? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Pooling and spatial information (ai-5/22). Goal: Compute max and mean pooling under an explicit window rule. Short practice: Can the max map reveal where the 4 originally appeared? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
